LAB 2 — AGILE BACKLOG CREATION & SPRINT SIMULATION

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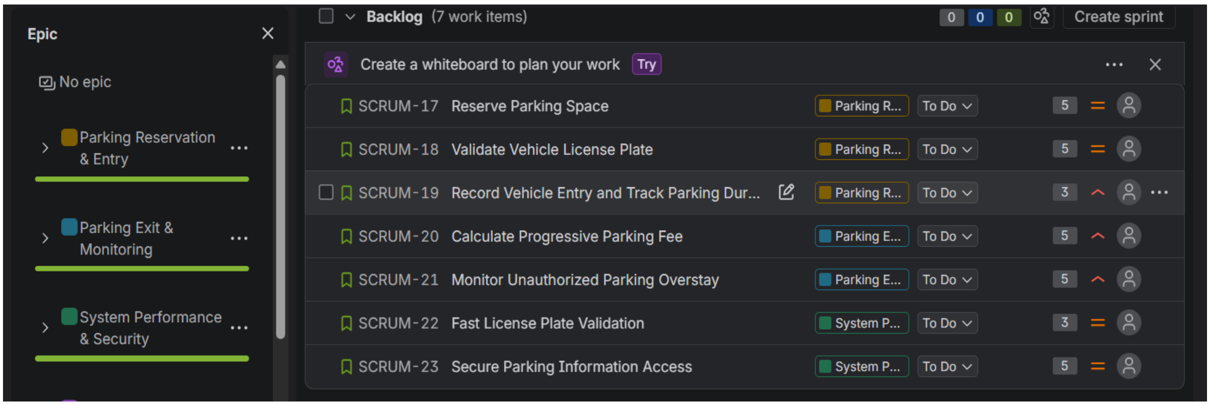
DATE : 01/09/2026 SEC : 5D

Problem Statement: Smart Parking Space Rental & License Validator

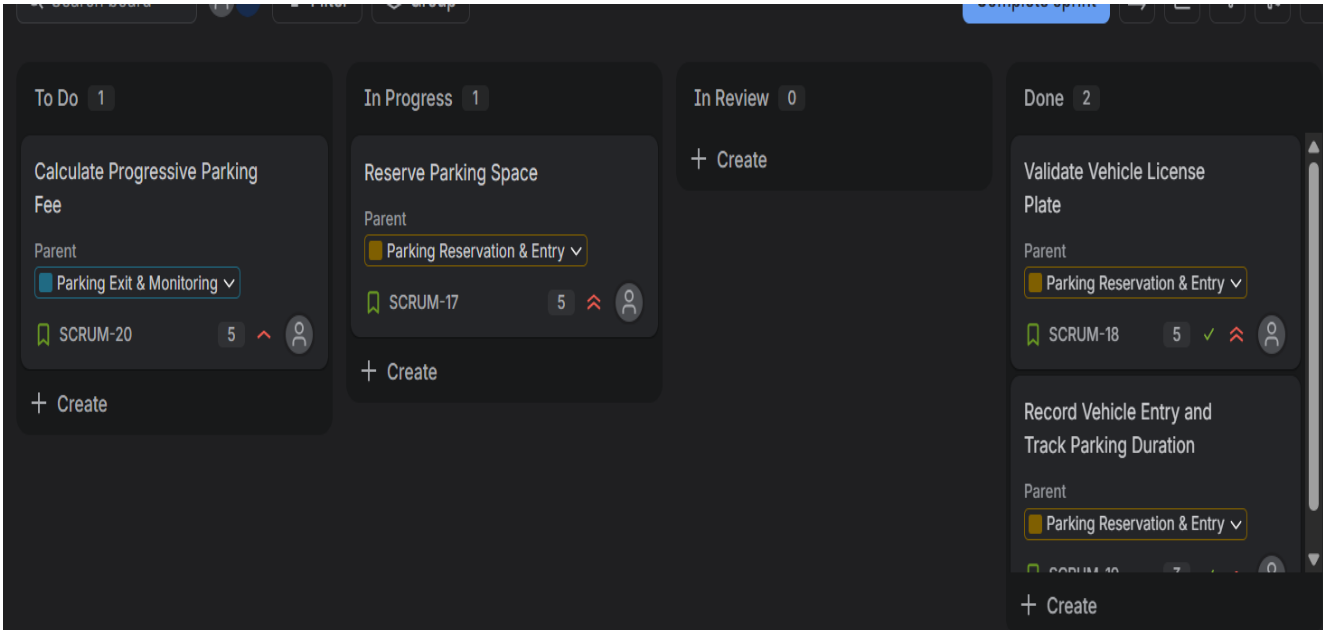
1. Epics and User Stories

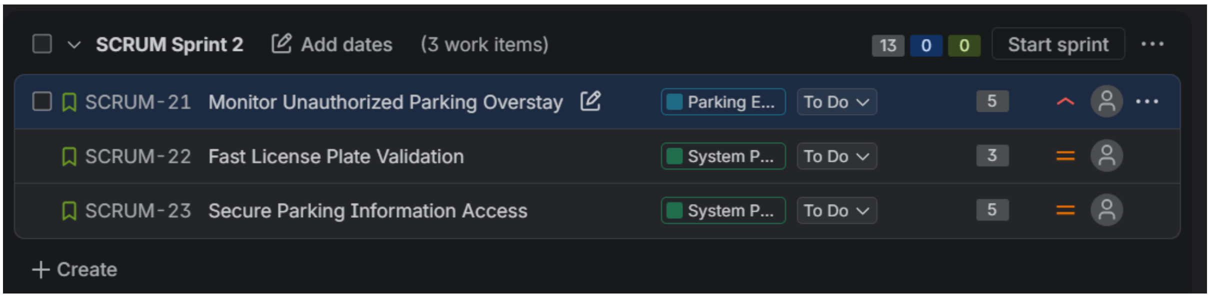


Sl. No.	Epic	User Story
1.	**Parking Reservation & Entry**	**Reserve Parking Space** As a vehicle owner, I want to reserve an available parking space so that I can secure a parking spot before arriving.
2.	**Parking Reservation & Entry**	**Validate Vehicle License Plate** As a parking warden, I want to validate the vehicle's license plate so that only authorized vehicles can enter.
3.	**Parking Reservation & Entry**	**Record Vehicle Entry and Track Parking Duration** As a parking warden, I want to record vehicle entry and track parking duration so that the parking time can be monitored accurately.
4.	**Parking Exit & Monitoring**	**Calculate Progressive Parking Fee** As a vehicle owner, I want the parking fee to be calculated

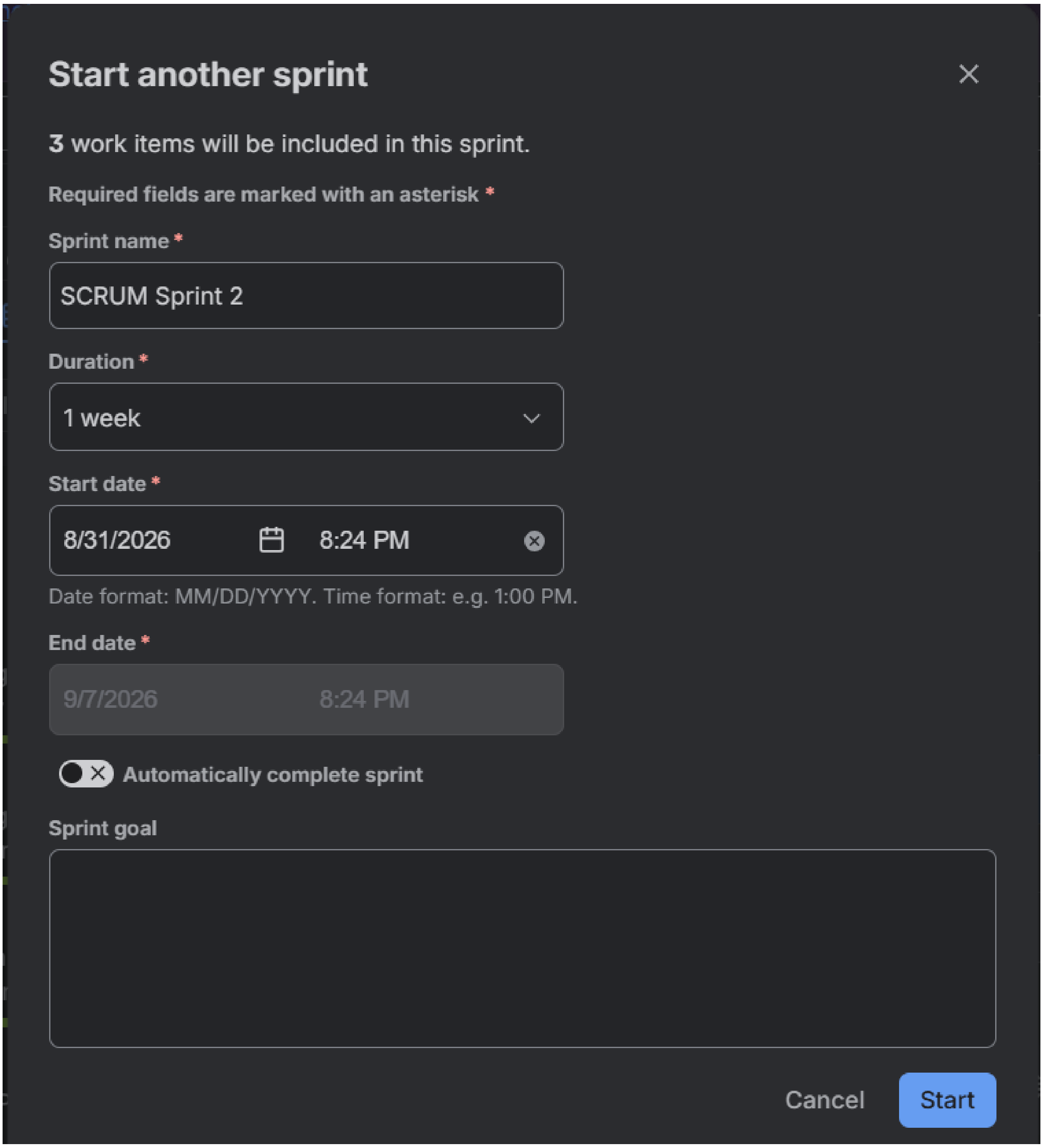
		based on my parking duration so that I can pay the correct amount.
5.	**Parking Exit & Monitoring**	**Monitor Unauthorized Parking Overstay** As a parking warden, I want to monitor vehicles that exceed their permitted parking time so that unauthorized overstays can be identified.
6.	**System Performance & Security**	**Fast License Plate Validation** As a parking warden, I want license plates to be validated quickly so that vehicle entry is not delayed.
7.	**System Performance & Security**	**Secure Parking** Information Access As a parking warden, I want parking information to be securely stored and accessed so that vehicle and reservation data are protected.

2. **Sprint board**
   Evidence for Sprint 1.

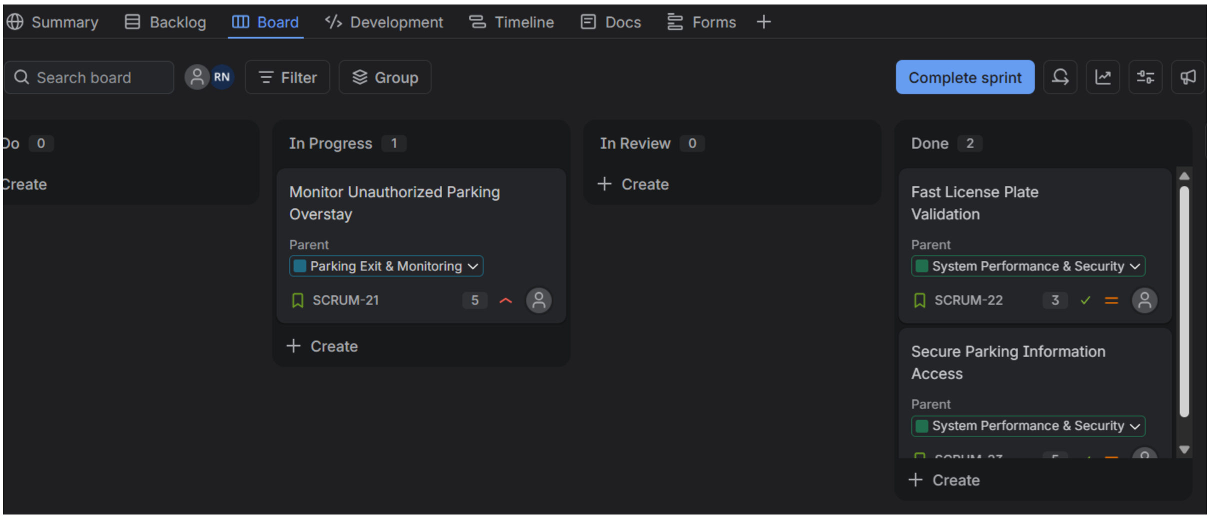




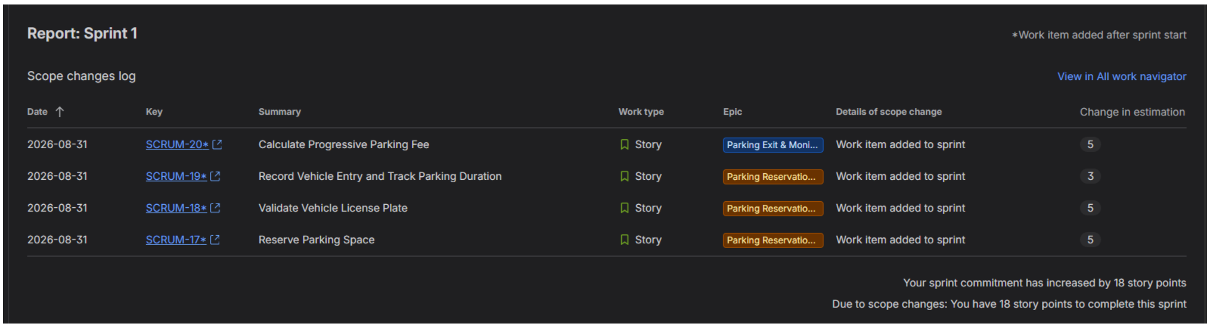
3. **<u>Sprint 2 , with its duration set to 1 week.</u>**



Evidence for Sprint 2.



4. Sprint 1 Burndown Chart



Report: Sprint 1

*Work item added after sprint start

Scope changes log

View in All work navigator

Date	Key	Summary	Work type	Epic	Details of scope change	Change in estimation
2026-08-31	SCRUM-20*	Calculate Progressive Parking Fee	Story	Parking Exit & Moni...	Work item added to sprint	5
2026-08-31	SCRUM-19*	Record Vehicle Entry and Track Parking Duration	Story	Parking Reservatio...	Work item added to sprint	3
2026-08-31	SCRUM-18*	Validate Vehicle License Plate	Story	Parking Reservatio...	Work item added to sprint	5
2026-08-31	SCRUM-17*	Reserve Parking Space	Story	Parking Reservatio...	Work item added to sprint	5

Your sprint commitment has increased by 18 story points

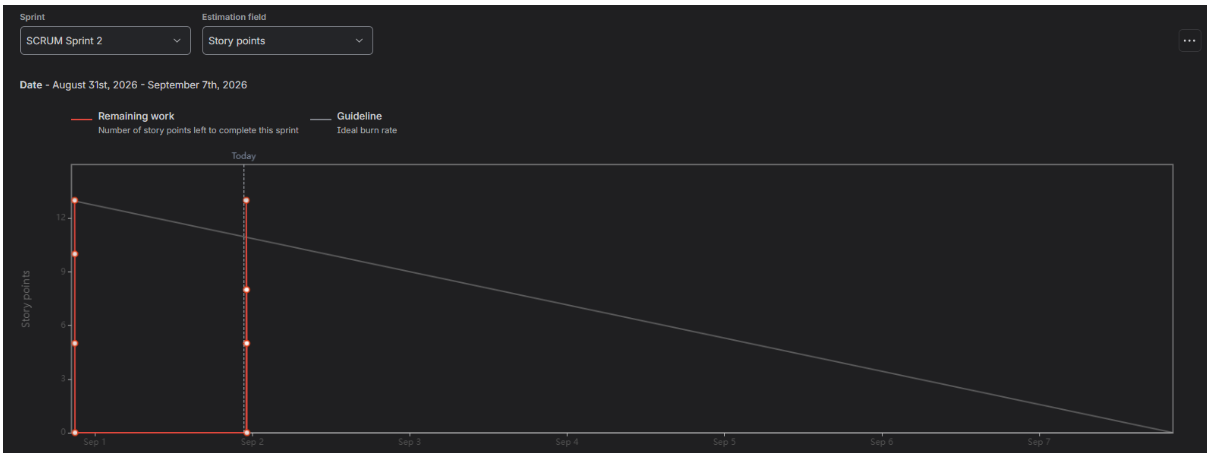
Due to scope changes: You have 18 story points to complete this sprint

Completed work items

View in All work navigator

Key	Summary	Work type	Epic	Status	Assignee	Story points
SCRUM-20	Calculate Progressive Parking Fee	Story	Parking Exit & Moni...	Done		5
SCRUM-19	Record Vehicle Entry and Track Parking Duration	Story	Parking Reservatio...	Done		3
SCRUM-18	Validate Vehicle License Plate	Story	Parking Reservatio...	Done		5
SCRUM-17	Reserve Parking Space	Story	Parking Reservatio...	Done		5

5. Sprint 2 Burndown Chart

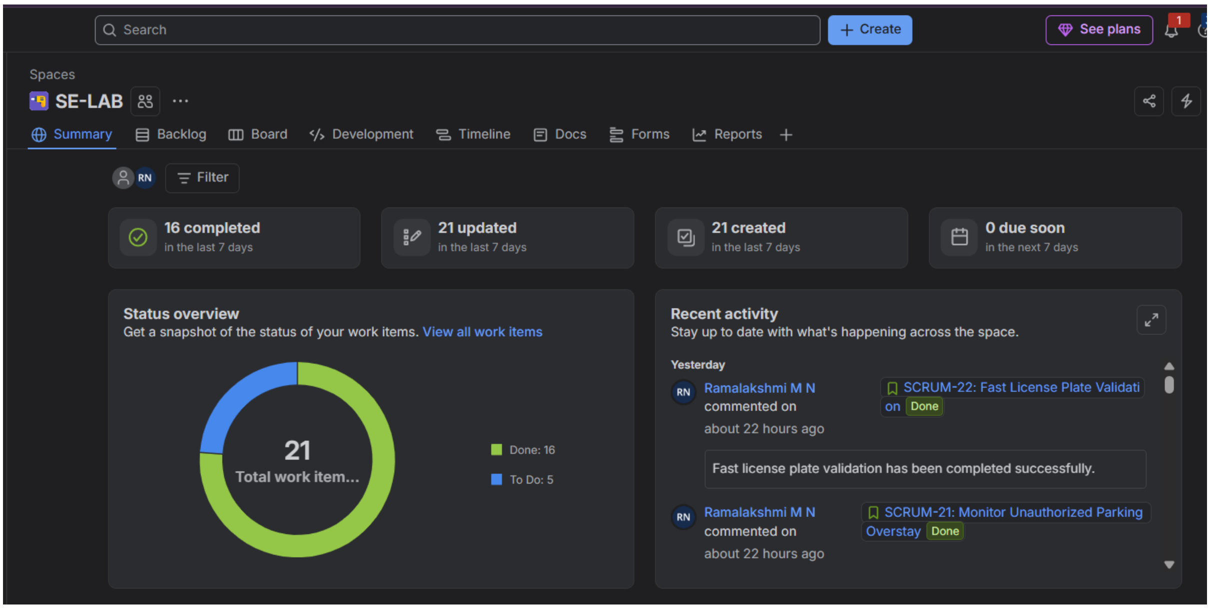


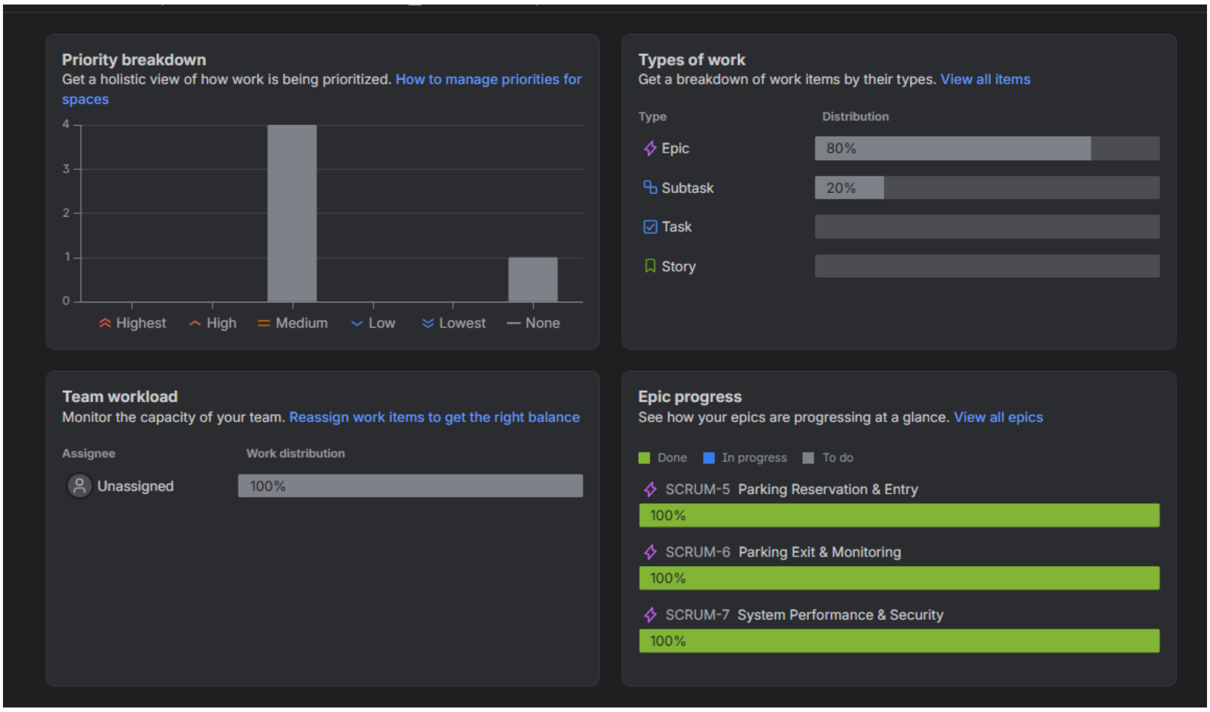
Completed work items

View in All work navigator

Key	Summary	Work type	Epic	Status	Assignee	Story points
SCRUM-21	Monitor Unauthorized Parking Overstay	Story	Parking Exit & Moni...	Done		5
SCRUM-22	Fast License Plate Validation	Story	System Performan...	Done		3
SCRUM-23	Secure Parking Information Access	Story	System Performan...	Done		5

6.SUMMARY:





7. Group Reflection

Reflection

This lab helped us understand how Agile and Scrum work using Jira. We learned how to create Epics and User Stories, assign priorities and Story Points, and organize the work into sprints. We also learned how to track stories from To Do → In Progress → Done and use the Burndown Chart to

check sprint progress. Overall, this lab gave us a better understanding of how Jira can be used to manage and track a project.

Q1. What did you learn from this lab?
Answer: We learned how to create and manage a backlog, plan sprints, and track the progress of User Stories using Jira.

Q2. Why are Story Points used?
Answer: Story Points help us estimate how much effort and complexity is involved in completing a User Story.

Q3. Why did we prioritize the User Stories?
Answer: We prioritized them so that the more important requirements could be completed first.

Q4. What did you learn from the two sprints?
Answer: We learned how to divide the work into smaller parts and track each User Story through To Do, In Progress, and Done.

Q5. What is the use of a Burndown Chart?
Answer: It shows how much work is completed and how much work is still remaining in a sprint.

Q6. What was the main challenge you faced?
Answer: Initially, understanding the Jira interface and organizing the stories into Epics and Sprints was a little confusing, but after using it, the process became easier.

Q7. What can be improved in future sprints?
Answer: We can improve estimation and sprint planning so that the right amount of work is selected for each sprint.

Q8. What was your main takeaway from this lab?
Answer: My main takeaway was understanding how Agile, Scrum, and Jira can be used together to organize and track software development work.